

Bar Models and Part-Whole Reasoning

Mission

Warm-up and worked example

Name: _____

Date: _____

PAGE 1 | WEEK 2, DAY 3

YOUR MISSION Represent unknown parts, wholes, and comparisons with bar models and equations.

SMART MOVE Estimate first, then use place value to check every trade.

Math Talk

The whole is 850 and one part is 327. What is the other part?

My idea:

New Words

addend	sum	difference
My meaning or picture:	My meaning or picture:	My meaning or picture:

Worked Example

Do this example with your teacher. Say what changes and what stays the same.

1. Draw a bar model for $\text{part} + \text{part} = \text{whole}$.
2. Draw a comparison model for "358 more than 467."

What the example shows: _____

Bar Models and Part-Whole Reasoning

Mission

Learn it and show your thinking

Name: _____

Date: _____

PAGE 2 | WEEK 2, DAY 3

YOUR MISSION Follow the learning path, then explain the big idea.

SMART MOVE Draw before asking for a rule.

[] **Step 1:** Draw a bar model for $\text{part} + \text{part} = \text{whole}$.

[] **Step 2:** Draw a comparison model for "358 more than 467."

[] **Step 3:** Label known and unknown quantities.

[] **Step 4:** Write an equation with a box or letter for the unknown.

[] **Step 5:** Solve and check the answer in the story.

My model, drawing, or organized work:

The big idea in my own words:

Bar Models and Part-Whole Reasoning

Mission

Practice lab

Name: _____

Date: _____

PAGE 3 | WEEK 2, DAY 3

YOUR MISSION Use a model, equation, or explanation for every task.

SMART MOVE Check whether your answer is reasonable.

1. Find an unknown part.

2. Find an unknown whole.

3. Solve a "more than" comparison.

4. Solve a "less than" comparison.

5. Match four equations to four bar models.

Choose one answer and prove it another way:

Bar Models and Part-Whole Reasoning

Mission

Challenge and final check

Name: _____

Date: _____

PAGE 4 | WEEK 2, DAY 3

YOUR MISSION Try an unfamiliar problem and defend your reasoning.

SMART MOVE A wrong first idea can still lead to a strong solution.

Challenge Mission

A number is 358 more than 467 and 129 less than another number. Find the other number.

My plan:

My solution and proof:

Final Check

Draw a bar model for $x + 275 = 902$.

☐ I can teach it

☐ I need practice

☐ I need help

Bar Models and Part-Whole Reasoning

Mission

Reusable math tool

Name: _____

Date: _____

PAGE 5 | WEEK 2, DAY 3

YOUR MISSION Use this page to draw, model, organize, and check.

SMART MOVE Estimate first, then use place value to check every trade.

Model and Record

Model	Equation	Check	Explain